Cybersecurity Isolated Lab Setup Report

Task 1: VirtualBox NAT Network Architecture, Kali Linux Deployment & Baseline Verification

Program:	Networkwalks Cybersecurity Training	**Role:**	Security Intern / Student	**Target Subnet:**	10.0.2.0/24	**Status:**	Completed & Verified

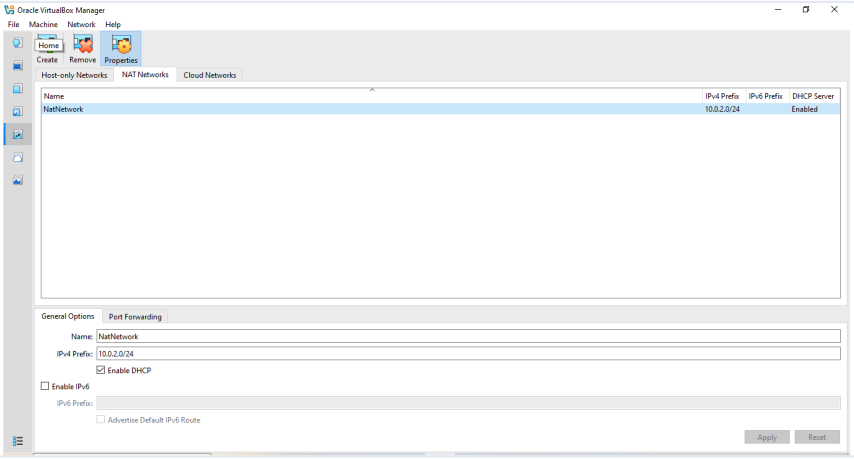
1. Executive Summary & Lab Objectives

Establishing an isolated and controlled virtualization environment is the foundational prerequisite for penetration testing, malware analysis, and defensive assessments. The objective of this assignment is to deploy Oracle VM VirtualBox, construct a dedicated **NAT Network** to allow internal VM-to-VM traffic while preserving host and Internet routing, bind Kali Linux to this adapter, and validate internal addressing, gateway connectivity, and external domain name resolution (DNS).

Environment Component	Configuration Parameter	Assigned Value	Functional Role
Hypervisor Platform	Oracle VM VirtualBox	v7.x (AMD64)	Isolated Type-2 Virtualization Host
Virtual Network	Custom NAT Network	`NatNetwork` (10.0.2.0/24)	Internal VM communication + outbound WAN access
Security Workstation	Kali Linux	2026.2-amd64 (RAM: 2GB+)	Primary offensive assessment platform
IP Allocation	DHCP Service Enabled	Assigned: `10.0.2.3/24`	Automatic addressing within private lab segment

2. Step 1: VirtualBox NAT Network Configuration

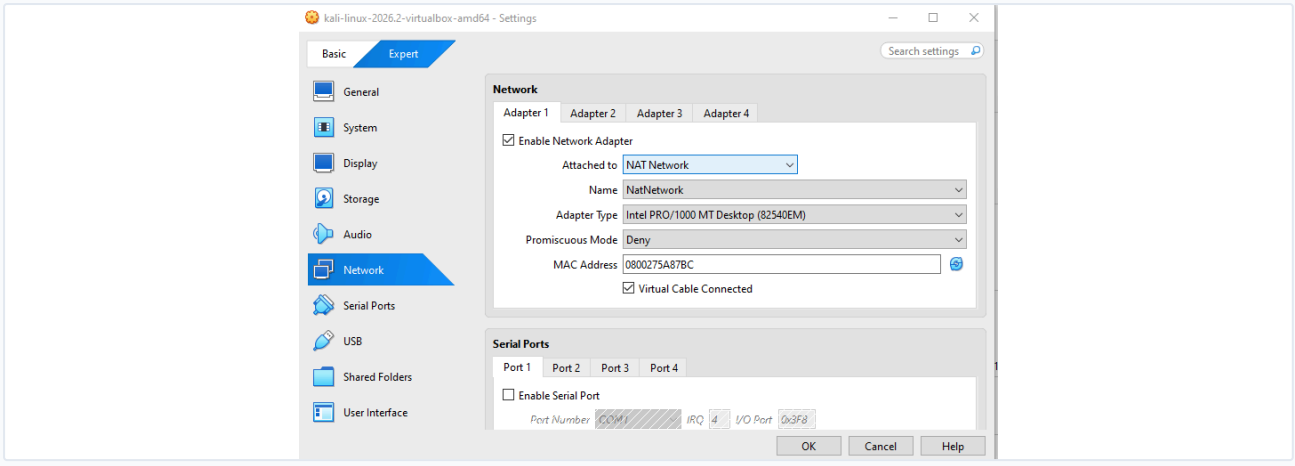
Standard default NAT provides isolated egress translation without inter-VM connectivity. By deploying a dedicated **NAT Network** under Global Network Tools, future victim machines (e.g., Metasploitable, Windows targets) will seamlessly communicate directly with the Kali adversary machine while preventing accidental network pollution to the local physical LAN.



VirtualBox Network Manager — NAT Network profile (NatNetwork) configured with 10.0.2.0/24 and active DHCP. **CONFIGURED**

3. Step 2: Virtual Machine Network Adapter Association

The Kali Linux virtual appliance was previously attached to a Bridged adapter. To isolate network traffic safely, the virtual machine network settings were reconfigured. **Adapter 1** was attached to **NAT Network**, targeting the newly instantiated network identifier **NatNetwork**. Promiscuous mode was left in Deny mode, maintaining strict host-isolation boundaries.

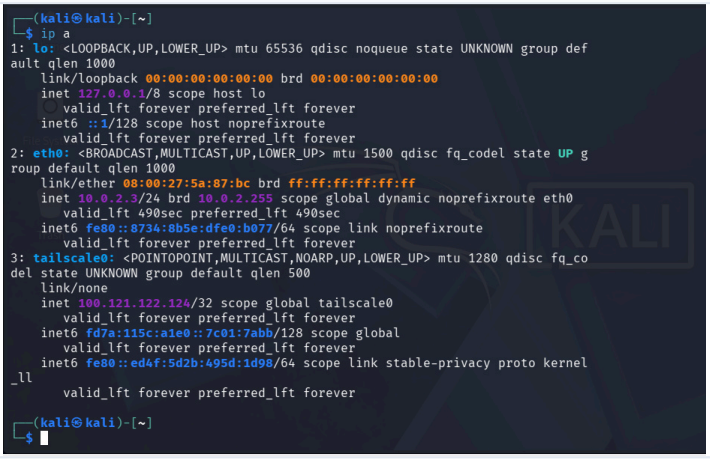


Kali Linux VM Settings — Adapter 1 mapped to NatNetwork with Virtual Cable Connected.

ASSOCIATED

4. Step 3: Kali Linux Interface & IP Addressing Verification

Upon booting the Kali virtual appliance, networking services were queried using the Linux `ip a` utility. Interface **eth0** successfully obtained an address lease from the VirtualBox internal DHCP server, leasing **10.0.2.3/24** with broadcast **10.0.2.255**, confirming correct subnet attachment.

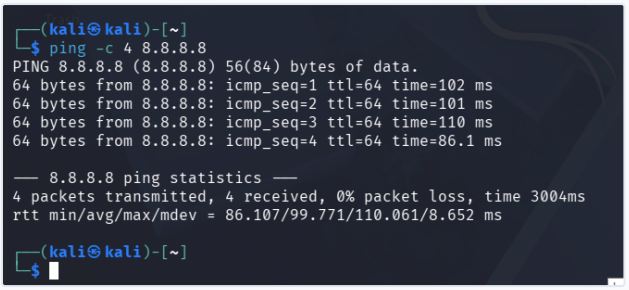


Terminal output showing eth0 state UP with assigned IP 10.0.2.3/24 under NatNetwork.

VERIFIED 10.0.2.3

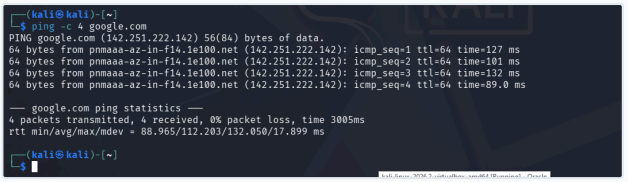
5. Step 4: Connectivity & DNS Resolution Testing

Network transport validation required testing both IP layer routing (ICMP echo requests to public DNS) and DNS query translation. Both diagnostics completed with **0% packet loss**, proving full WAN egress capability through the hypervisor's translation layer.



ICMP ping to 8.8.8.8 (0% loss, ~99ms latency).

WAN REACHABLE

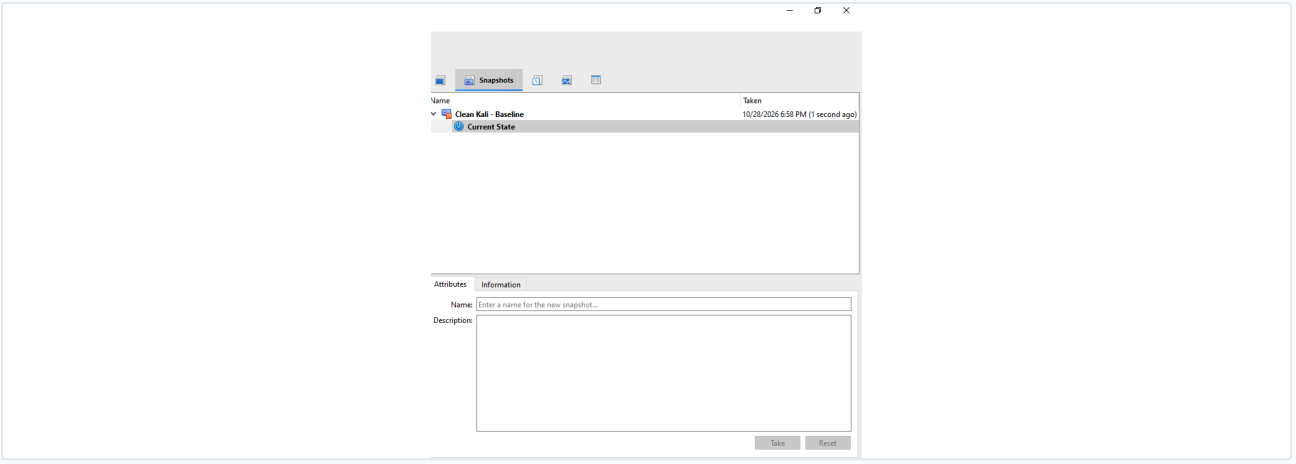


DNS name resolution test to google.com (0% loss).

DNS RESOLVED

6. Step 5: Clean Baseline Snapshot Creation

To safeguard the operating environment against misconfigurations during future exploits, scans, or software installations, the system was cleanly powered down and an initial state snapshot was captured: **Clean Kali - Baseline**.



VirtualBox Manager — Clean Kali - Baseline state snapshot saved successfully.

SNAPSHOT SAVED